

06 Multilinear Regression Home Price

Markdown Cell 1

Experiment 6 — MultiLinear Regression on Home Price Dataset

Source: Multilinear Regression PDF

Implementation of MultiLinear Regression on the Home Price Dataset.

Code Cell 2

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn import linear_model

df=pd.read_csv('home_price.csv')
df
```

Code Cell 3

```
import math
median_bedrooms = math.floor(df.bedrooms.median())
median_bedrooms
```

Code Cell 4

```
df.bedrooms = df.bedrooms.fillna(median_bedrooms)
df
```

Code Cell 5

```
from sklearn.model_selection import train_test_split

x_train,x_test,y_train,y_test = train_test_split(
    df[['area','bedrooms','age']],
    df.price,
    test_size=0.20,
    shuffle=False
)
```

Code Cell 6

```
reg = linear_model.LinearRegression()
reg.fit(x_train, y_train)
```

Code Cell 7

```
reg.coef_
```

Code Cell 8

```
reg.intercept_
```

Code Cell 9

```
reg.score(x_test, y_test)
```

Code Cell 10

```
reg.predict([[3000,3,40]])
```